

2. (2.75 points)

Given the following information for a workers compensation book written with a per-occurrence deductible of \$500,000:

Fitted Limited Severity		
Limit	36 months	Ultimate
Unlimited	\$5,000	\$9,000
\$500,000	\$4,500	\$6,600

	36 months	Ultimate
Observed Deductible Loss	\$1,000,000	-
Observed Excess Loss	\$275,000	-
Modeled Excess Age-to-Ultimate Factor	3.00	-
Insurance Charge Ratio	0.20	0.30

- Expected claims volume = 1,000 claims.

a. (2.25 points)

Assess the reasonability of the modeled excess IBNR compared to the loss ratio approach.

b. (0.5 point)

Given the following:

Volume Weighted Claim Count Age-to-Age Development Factors as of (months)				
12-24	24-36	36-48	48-60	60-72
1.50	1.25	1.15	1.05	1.00

Describe an issue with using limited severity data at 36 months.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 2	
TOTAL POINT VALUE: 2.75	LEARNING OBJECTIVE(S): A4
SAMPLE ANSWERS	
Part a: 2.25 points	
<i>Sample responses are separated into the three main pieces of the question:</i>	
Modeled XS IBNR:	
<u>Sample 1</u>	
XS IBNR = Reported XS Loss × Modeled XS LDF – Reported XS LDF = 275K × 3 – 275K = 550K	
<u>Sample 2</u>	
XS IBNR = Reported XS Loss × Derived XS LDF – Reported XS LDF = 275K × 4.806 – 275K = 1,045K	
$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}} = 1.80 \times (1 - 0.733) / (1 - 0.900) = 4.806$	
$R_{ult}^{500} = 6,600 / 9,000 = 0.733$	
$R_{36}^{500} = 4,500 / 5,000 = 0.900$	
$LDF_{36-ult}^{unlim} = 9,000 / 5,000 = 1.80$	
IBNR from LR Approach:	
<u>Sample 1</u>	
$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}} = 1.80 \times (1 - 0.733) / (1 - 0.900) = 4.806$	
$R_{ult}^{500} = 6,600 / 9,000 = 0.733$	
$R_{36}^{500} = 4,500 / 5,000 = 0.900$	
$LDF_{36-ult}^{unlim} = 9,000 / 5,000 = 1.80$	
XS Ult Loss = XS Sev × Counts = (9,000 – 6,600) × 1,000 = 2.4M	
XS IBNR = 2.4M × (1 – 1 / 4.806) = 1,900,624	
<u>Sample 2</u>	
Ult XS = (Ult Total Sev – Ult Limited Sev) × Counts = (9,000 – 6,600) × 1,000 = 2.4M	
XS@36 mo = (Total Sev@36 mo – Limited Sev@36 mo) × Counts = (5,000 – 4,500) × 1,000 = 0.5M	
XS IBNR = 2.4M – 0.5M = 1.9M	
<u>Sample 3</u>	
Ult XS = (Ult Total Sev – Ult Limited Sev) × Counts = (9,000 – 6,600) × 1,000 = 2.4M	
XS IBNR = Ult XS – Rept XS = 2.4M – 275K = 2,125K	
<u>Sample 4</u>	
Ult XS = (Ult Total Sev – Ult Limited Sev) × Counts = (9,000 – 6,600) × 1,000 = 2.4M	
XS IBNR = Ult XS × (1 – 1 / Modeled XS LDF) = 2.4M × (1 – 1/3.00) = 1.6M	

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Sample 5

LR approach = $P \cdot E \cdot \chi$ - Rept XS

Ult XS = Ult Total Sev $\times$ Counts $\times$ Ins Charge@Ult = $9,000 \times 1,000 \times 0.30 = 2.7M$

Rept XS = Total Report Loss $\times$ Ins Charge@36 mo = $(1M + 275K) \times 0.20 = 255K$

XS IBNR = $2.7M - 255K = 2,445K$

Sample 6

LR approach = $P \cdot E \cdot \chi$ - Rept XS

Derived Unlim LDF = Ult Total Sev / Total Sev@36 mo = $9K / 5K = 1.8$

Ult XS = Total Reported Loss $\times$ Derived Unlim LDF $\times$ Ins Charge@Ult = $(1M + 275K) \times 1.80 \times 0.30 = 688.5K$

Rept XS = Total Report Loss $\times$ Ins Charge@36 mo = $(1M + 275K) \times 0.20 = 255K$

XS IBNR = Ult XS - Rept XS = $688.5K - 255K = 413.5K$

Sample 7

LR approach = $P \cdot E \cdot \chi$ - Rept XS

$$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}}$$

$$3.00 = LDF_{36-ult}^{unlim}$$

$$R_{ult}^{500} = 6,600 / 9,000 = 0.733$$

$$R_{36}^{500} = 4,500 / 5,000 = 0.900$$

$$LDF_{36-ult}^{unlim} = 3.00 \times 0.1 / 0.267 = 1.125$$

Ult XS = Total Reported Loss $\times$ Derived Unlim LDF $\times$ Ins Charge@Ult = $(1M + 275K) \times 1.125 \times 0.30 = 430.3125K$

Rept XS = Total Report Loss $\times$ Ins Charge@36 mo = $(1M + 275K) \times 0.20 = 255K$

XS IBNR = Ult XS - Rept XS = $430.3125K - 255K = 175.3125K$

Sample 8

LR approach = $P \cdot E \cdot \chi$ - Rept XS

$$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}}$$

$$3.00 = LDF_{36-ult}^{unlim}$$

$$R_{ult}^{500} = 6,600 / 9,000 = 0.733$$

$$R_{36}^{500} = 4,500 / 5,000 = 0.900$$

$$LDF_{36-ult}^{unlim} = 3.00 \times 0.1 / 0.267 = 1.125$$

Ult XS = Total Reported Loss $\times$ Derived Unlim LDF $\times$ Ins Charge@Ult = $(1M + 275K) \times 1.125 \times 0.30 = 430.3125K$

Rept XS = 275K (Given)

XS IBNR = Ult XS - Rept XS = $430.3125K - 275K = 155.3125K$

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Sample 9

LR approach = $P \cdot E \cdot \chi$ - Rept XS

$$LDF_{36-ult}^{unlim} = 9K / 5K = 1.80$$

$$R_{ult}^{500} = 6,600 / 9,000 = 0.733$$

$$R_{36}^{500} = 4,500 / 5,000 = 0.900$$

$$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}}$$

$$LDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{R_{ult}^{500}}{R_{36}^{500}}$$

$$XSLDF_{36-ult} = 1.80 \times (1 - 0.733) / (1 - 0.900) = 4.80$$

$$LDF_{36-ult} = 1.80 \times 0.733 / 0.900 = 1.467$$

$$\text{Ult XS Loss} = 275K \times 4.8 = 1,320K$$

$$\text{Ult Lim Loss} = 1M \times 1.467 = 1,467K$$

$$\text{Ult Total Loss} = \text{Ult XS Loss} + \text{Ult Lim Loss} = 1,320K + 1,467K = 2,787K$$

$$\text{Ult XS} = \text{Ult Total Loss} \times \text{Ins Charge@Ult} = 2,787K \times 0.30 = 836K$$

$$\text{Rept XS} = (1,000K + 275K) \times 0.20 = 255K$$

$$\text{XS IBNR} = \text{Ult XS} - \text{Rept XS} = 836K - 275K = 561K$$

Sample 10

LR approach = $P \cdot E \cdot \chi$ - Rept XS

$$\text{Excess Ratio} = 275K / 1,275K = 0.216$$

$$\text{Ult XS} = \text{Ult Total Sev} \times \text{Counts} \times \text{Derived Excess Ratio} = 9,000 \times 1,000 \times 0.216 = 1,941K$$

$$\text{Rept XS} = 275K \text{ (Given)}$$

$$\text{XS IBNR} = \text{Ult XS} - \text{Rept XS} = 1,941K - 275K = 1,666K$$

Sample 11

LR approach = $P \cdot E \cdot \chi$ - Rept XS

$$LDF_{36-ult}^{unlim} = 9K / 5K = 1.80$$

$$R_{ult}^{500} = 6,600 / 9,000 = 0.733$$

$$R_{36}^{500} = 4,500 / 5,000 = 0.900$$

$$LDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{R_{ult}^{500}}{R_{36}^{500}}$$

$$LDF_{36-ult} = 1.80 \times 0.733 / 0.900 = 1.467$$

$$\text{Ult XS Loss} = \text{Rept Lim Loss} \times \text{Derived Limited LDF} \times \text{Insurance Charge Ult} = 1M \times 1.467 \times 0.30 = 440K$$

$$\text{Rept XS} = \text{Rept Lim Loss} \times \text{Insurance Charge@36 mo} = 1,000K \times 0.20 = 200K$$

$$\text{XS IBNR} = \text{Ult XS} - \text{Rept XS} = 440K - 200K = 240K$$

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Sample 12

LR approach = $P \cdot E \cdot \chi$

Exp XS@36 mo = Total Sev@36 mo $\times$ Counts $\times$ Insurance Charge@36 mo = $5,000 \times 1,000 \times 0.20 = 1\text{M}$

Exp Ultimate XS = Ultimate Total Sev $\times$ Counts $\times$ Insurance Charge Ult = $9,000 \times 1,000 \times 0.30 = 2.7\text{M}$

XS IBNR = Exp Ult XS – Exp XS@36 mo = $2.7\text{M} - 1\text{M} = 1.7\text{M}$

Sample 13

LR approach = $P \cdot E \cdot \chi$

Exp XS@36 mo = Limited Sev@36 mo $\times$ Counts $\times$ Insurance Charge@36 mo = $4,500 \times 1,000 \times 0.20 = 900\text{K}$

Exp Ultimate XS = Ultimate Limited Sev $\times$ Counts $\times$ Insurance Charge Ult = $6,600 \times 1,000 \times 0.30 = 1,980\text{K}$

XS IBNR = Exp Ult XS – Exp XS@36 mo = $1,980\text{K} - 900\text{K} = 1,080\text{K}$

Sample 14

Ult XS = Ult Total Sev $\times$ Counts $\times$ Ins Charge @ Ult = $9,000 \times 1,000 \times 0.3 = 2.7\text{M}$

XS IBNR = Ult XS – Rept XS = $2.7\text{M} - 275\text{K} = 2,425\text{K}$

Sample 15

$$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}}$$

$$XSLDF_{36-ult} = 1.8 \times (1 - 0.733) / (1 - 0.9) = 4.8$$

$$XS \text{ IBNR} = 275\text{K} \times (4.8 - 1) = 1,045\text{K}$$

Sample 16

XS Ultimate = Ultimate Total Loss $\times$ XS Ratio + Ultimate Total Loss $\times$ (1 – XS Ratio) $\times$ Aggregate Ratio

$$XS \text{ Ultimate} = 9,000 \times 1,000 \times (1 - 6,600 / 9,000) + 9,000 \times 1,000 \times (6,600 / 9,000) \times 0.3 = 4.38\text{M}$$

$$XS \text{ IBNR} = XS \text{ Ultimate} - \text{Rept XS} = 4.38\text{M} - 275\text{K} = 4.105\text{M}$$

Sample 17

275K = Expected Loss@36 mo $\times$ Insurance Charge@36 mo

$$\text{Expected Loss@36 mo} = 275\text{K} / 0.2 = 1,375\text{K}$$

$$\text{Ultimate Total Loss} = 1,375\text{K} \times \text{Unlimited LDF} = 1,375\text{K} \times 9,000 / 5,000 = 2,475\text{K}$$

$$\text{Ultimate XS Loss} = \text{Ultimate Total Loss} \times \text{Insurance Charge Ult} = 2,475\text{K} \times 0.3 = 742.5\text{K}$$

$$XS \text{ IBNR} = \text{Ultimate XS Loss} - \text{Rept XS} = 742.5\text{K} - 275\text{K} = 467.5\text{K}$$

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Sample 18

LR approach = $P \cdot E \cdot \chi - \text{Rept XS}$

$$XSLDF_{36-ult} = LDF_{36-ult}^{unlim} \times \frac{1 - R_{ult}^{500}}{1 - R_{36}^{500}}$$

$$3.00 = LDF_{36-ult}^{unlim}$$

$$R_{ult}^{500} = 6,600 / 9,000 = 0.733$$

$$R_{36}^{500} = 4,500 / 5,000 = 0.900$$

$$LDF_{36-ult}^{unlim} = 3.00 \times 0.1 / 0.267 = 1.125$$

$$\text{Ultimate Total Loss} = (1,000K + 275K) \times 1.125 = 1,434.375K$$

$$\text{XS Ratio} = 1 - 6,600 / 9,000 = 0.267$$

$$\text{Ultimate XS Loss} = 1,434.375K \times 0.267 = 382.5K$$

$$\text{XS IBNR} = \text{Ultimate XS Loss} - \text{Rept XS} = 382.5K - 275K = 107.5K$$

Sample 19

LR approach = $P \cdot E \cdot \chi - \text{Rept XS}$

$$\text{Ultimate XS Loss} = \text{Ultimate Limited Sev} \times \text{Counts} \times \text{Insurance Charge} = 6,600 \times 1,000 \times 0.3 = 1,980K$$

$$\text{XS IBNR} = \text{Ultimate XS Loss} - \text{Rept XS} = 1,980K - 275K = 1,705K$$

Sample 20

$$\text{Ultimate Limited Loss} = 1,000K \times \text{Derived Limited LDF} = 1,000K \times (\text{Ultimate Limited Sev} / \text{Limited Sev@36 mo}) = 1,000K \times (6,600 / 4,500) = 1,466.67K$$

$$\text{Ultimate XS Loss} = 275K \times \text{Derived XS LDF} = 275K \times (\text{Ultimate XS Sev} / \text{Limited Sev@36 mo}) = 275K \times [(9,000 - 6,600) / (5,000 - 4,500)] = 275K \times 4.8 = 1.32M$$

$$\text{Ultimate Total Loss} = \text{Ultimate Limited Loss} + \text{Ultimate XS Loss} = 1.467M + 1.32M = 2.787M$$

$$\text{Ultimate XS Loss} = \text{Ultimate Total Loss} \times \text{Insurance Charge} = 2.787M \times 0.3 = 836K$$

$$\text{Rept XS} = \text{Total Loss@36 mo} \times \text{Insurance Charge@36 mo} = (1,000K + 275K) \times 0.2 = 255K$$

$$\text{XS IBNR} = \text{Ultimate XS Loss} - \text{Rept XS} = 836K - 255K = 581K$$

Assessment of 2 methods

Sample responses:

- The modeled XS IBNR is less reasonable compared to the LR approach. There may be imperfect data or uncertainty volatile in the actual loss experience.
- The modeled XS IBNR is higher than the LR approach because XS LDFs tend to be highly leveraged, especially in early maturities. And the LR approach doesn't take into consideration of actual experience and uses industry factors, which may not be appropriate. Using credibility approach of the two methods might be more appropriate.
- Modeled XS IBNR is much higher than calculated using LR method. This may be due to the highly leveraged XS LDF that is applied. LR approach is more stable than direct development.
- LR approach IBNR is much higher than modeled. This is because XS LDF is very volatile and

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uncertainty in estimation is high. Also we may have little reported XS loss at early maturity for WC which is long tail and subject to latent claims.

Part b: 0.5 point

Sample responses:

- Since the number of claims is expected to go up (ATA of 1.15 and 1.05), that will affect the severity unlimited and limited in different ways.
- By using the limited severity at 36 month, we are assuming that all claims have been reported by 36 month. Since there is still claim count development after 36 month, this violates the assumption and it is not proper to use the limited severity at 36 month.
- Since not all losses are reported at 36 months, the 36 months severity may be overstated, there are more claims expected which may be small.
- The issue is that the expected claims at age 36 is less than 1,000, only $1,000 / (1.15 \times 1.05) = 828$ are reported, likely those unreported are the most complex and severe. The unreported counts will have higher severity than the given severities at ultimate. Using the given severities will understate the IBNR.
- A number of claims have yet to be reported based on the development factors given. Claims that exceeded the deductible are likely to do so after longer periods of time, i.e. there is more development in the excess layer than in the deductible layer.

EXAMINER'S REPORT

Candidates were expected to apply Direct development method and Loss Ratio method (specifically Table M) to calculate the Excess layer IBNR, and assess the appropriateness of using the methods in a long-tailed line such as WC.

Candidates were also expected to understand the shortcomings of the Loss Ratio approach when claim counts are still developing.

Part a

Candidates were expected to calculate the XS IBNR using the Direct Development approach and the LR approach with the Table M (insurance charge) provided (or an alternative approach that calculates valid XS Ultimate Loss and Reported XS Loss), and then determine the most appropriate method to use in the given situation with logical support.

Common mistakes for LR and Modeled IBNR calculations included:

- Only calculating the ultimate loss without calculating the IBNR.
- Treating the ultimate loss as IBNR.
- Applying XS ultimate loss and reported loss to the insurance charge.
- Applying the wrong formula to get the limited/XS LDF when using the severity relativities (e.g., Limited LDF = Unlimited LDF $\times (1 - R_{Ult}) / (1 - R_{36})$, when it should be XS LDF).
- Applying both insurance charges (36 month and ultimate) to the ultimate loss.
- Not applying the insurance charge to the ultimate unlimited/limited loss.
- Applying Limited LDF to the total loss or applying the Unlimited LDF to the limited loss.

Common mistakes for the assessment of the 2 methods included:

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- Only stating that one method is lower/higher than the other method without providing valid reasoning, since numerical support alone was not sufficient (e.g., LR approach is higher than the Modeled approach, and therefore Modeled approach is not reasonable).

Part b

Candidates were expected to know that using counts that are still developing will distort the limited severity calculations or the IBNR calculations.

Common mistakes included:

- Calculating the XS/Limited counts LDF using the severity relativities even though too little information was given, and not tying back to the claim count development's impact on the severity calculation.
- Mentioning other valid issues of the LR approach without tying back to the claim count development, such as adjusting for inflation or different development patterns between excess and limited layer.